TIMER

Presented by

MD. NAIMUR RAHMAN — 2111076144
DISHAN IFTE NABIL — 2111076132
A.S.M SABBIR AHMED — 2110676154

What is a Timer?

A hardware peripheral for accurate time control

- A timer is a hardware module inside a microcontroller used to measure and control time.
- It counts clock pulses to create accurate time intervals.
- Timers can generate delays, periodic events, and interrupts.
- Example: blinking an LED every 1 second.

Timer vs Counter

Both count, but their input source is different

TIMER
- Uses the microcontroller’s internal clock.
- Example: generate a 1-second delay.

COUNTER
- Counts external events or pulses.
- Example: count how many times a button is pressed.

Basic Working Principle

How a timer produces a time event

CLOCK
↓
PRESCALER (PSC)
↓
COUNTER (CNT)
↓
Auto Reload Register (ARR)
↓
OVERFLOW / EVENT

1. Clock

Every timer needs a clock.

STM32 timers receive clock from RCC.

Suppose:

Timer Clock = 84 MHz

That means: 84 million counts every second.

Too fast!

So we reduce it.

2. Prescaler

Prescaler divides the clock.

Example:

Clock = 84 MHz
Prescaler = 84

Then,
84 MHz ÷ 84 = 1 MHz

Now,
Timer counts 1 million times per second.

3. Counter (CNT)

Counter stores current value.

0
1
2
3
4
5
...
Each clock pulse
Counter++

4. Auto Reload Register (ARR)

ARR tells the timer:

"When should I restart?"

Example:

ARR = 999

Counter
0
1
2
...
998
999

Next pulse
↓
Overflow
↓
Counter becomes 0 again.

The timer counter frequency is:

f_timer = f_clock / (PSC + 1)

Example:

Clock = 84 MHz
PSC = 83

84 MHz ÷ (83 + 1)
= 1 MHz

So: Counter increases 1 million times per second.

Numerical Example: 1-Second Timer

Using PSC and ARR to create a 1-second period

Clock = 72 MHz
PSC = 7199
CNT Clock = 10 kHz
ARR = 9999

1 Second → EVENT

Step 1: fCNT = 72,000,000 ÷ (7,199 + 1) = 10,000 Hz

Step 2: Period = (ARR + 1) ÷ fCNT = 10,000 ÷ 10,000 = 1 second

Result: So the timer generates an update event every 1 second.

Application: LED Blinking

A simple real-world timer example

SET (Clock + Prescaler)

↓

START (Start Timer)

↓

WAIT (CNT reaches ARR)

↓

TOGGLE (Toggle LED)

↓

REPEAT

Result: The LED turns ON and OFF at a fixed time interval.

Important Timer Registers

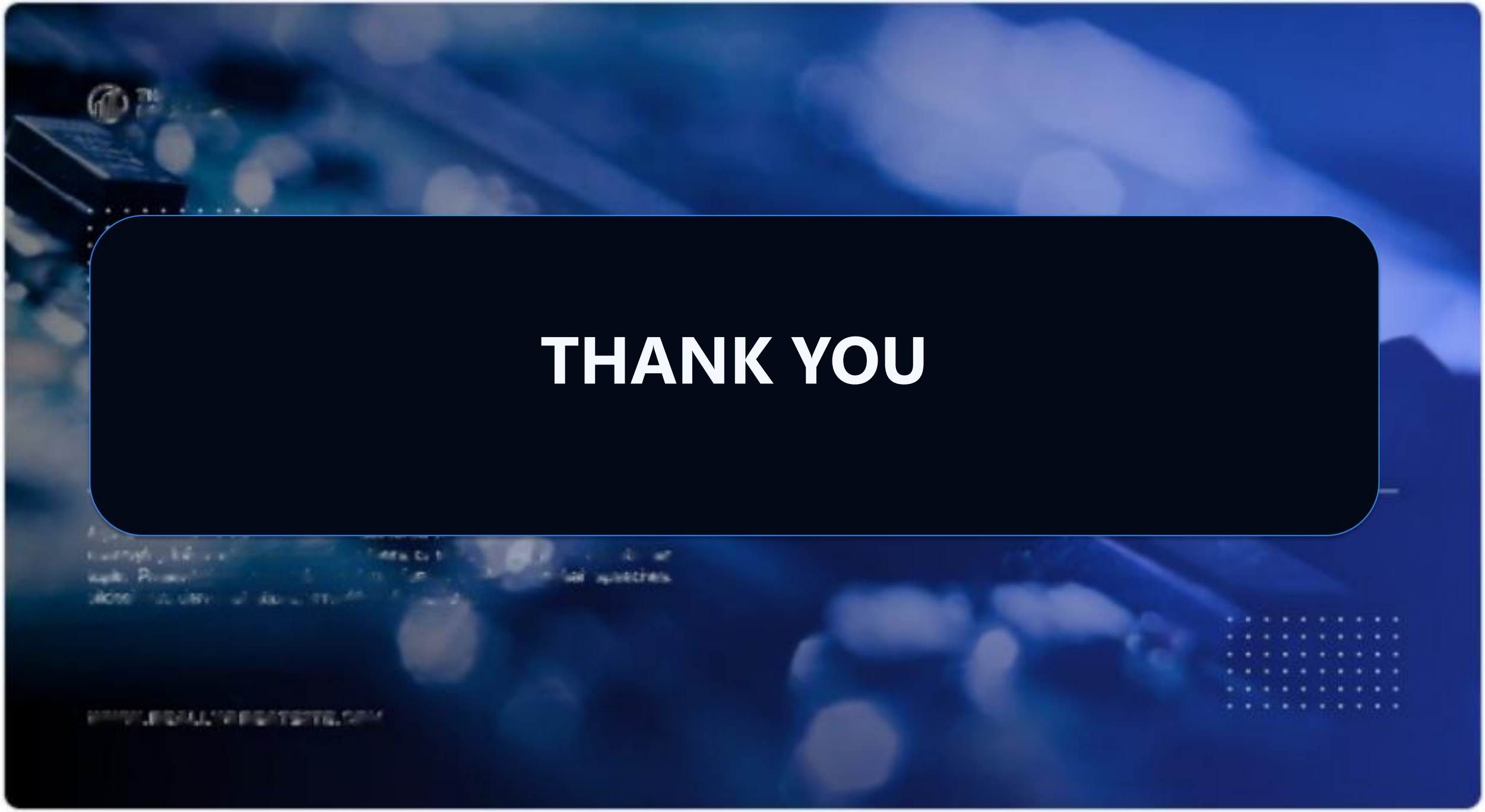
Typical registers used to configure a timer

Prescaler — Divides input clock.
Counter register — Current counter value.
Auto-Reload — Sets counter limit / period.
Control — Starts and configures timer.
Status — Shows timer events.

Conclusion

Why timers matter in embedded systems

- Timer is an essential microcontroller peripheral.
- It measures and controls time accurately.
- It can generate delays, interrupts, periodic signals and PWM.
- Timers are widely used in embedded and real-time applications.


THANK YOU